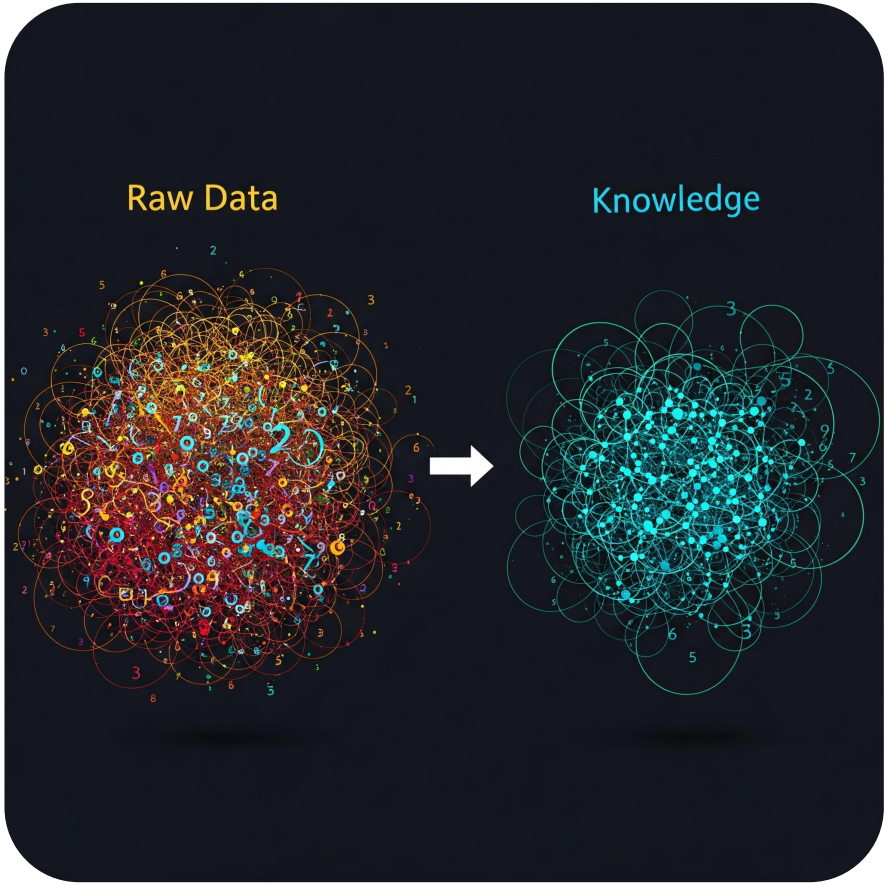
PORA ACADEMY

Data Science

Introduction to Data Science

Data and Data Science Defined

- Data: Raw facts, figures, or information.
- Data can be structured (e.g., sales figures) or unstructured (e.g., customer feedback).
- Data Science: The process of extracting knowledge and insights from data.
- Data Analysis: The process of inspecting, cleaning, transforming, and modeling data.



Understanding Data

Data refers to raw facts, figures, or information that can be collected, analyzed, and interpreted.

Data can be structured and unstructured.

- Examples of structured data: Sales figures,
- Examples of unstructured data: Customer feedback, social media posts.

***Discussion**: Mention the different data you have come across this week.*

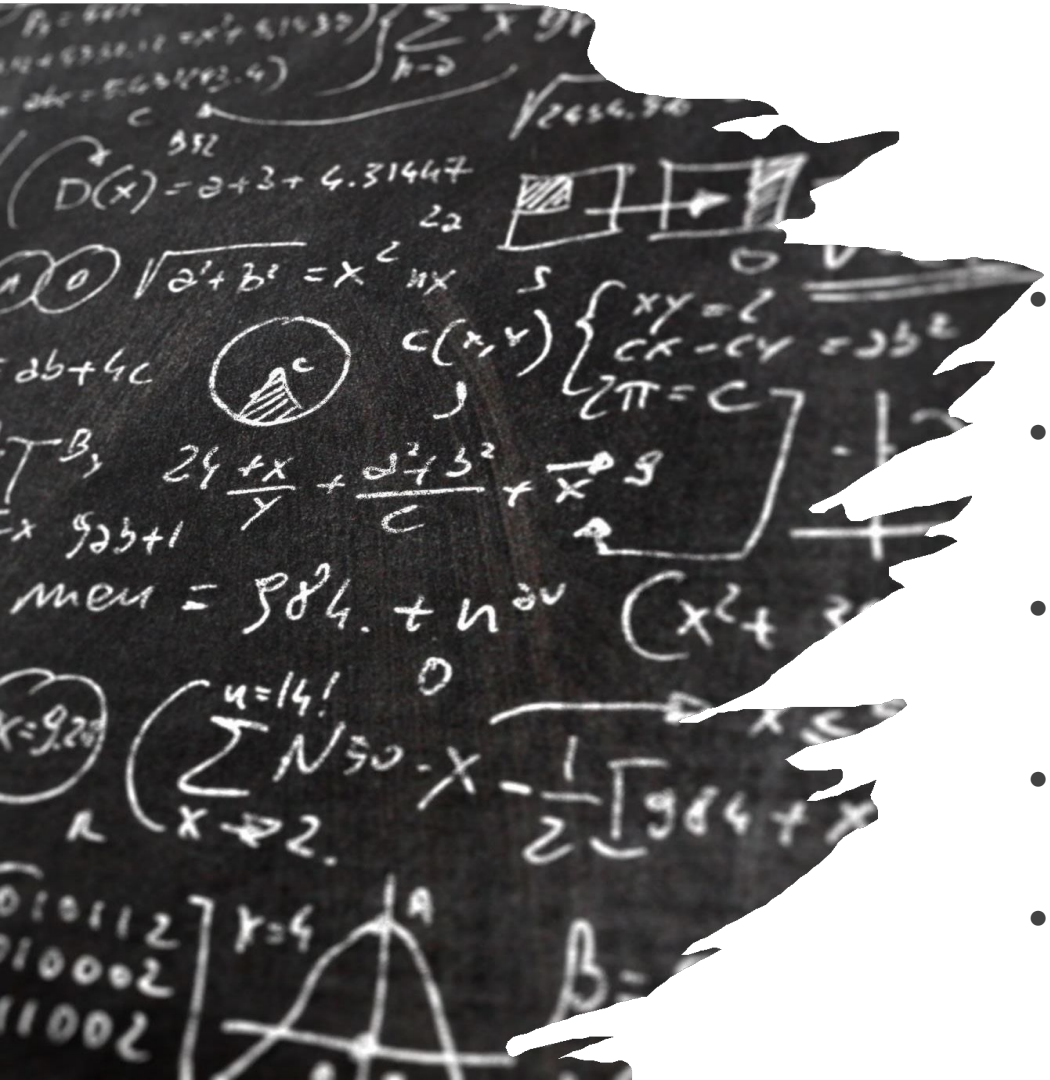
Facts about Data

- Over 90% of all the data in the world was created in the past 4 years.
- Less than 0.5% of all data we create is ever used or analyzed.
- If you burned all of the data created in just one day onto DVDs, you could stack them on top of each other and reach the moon – twice.

Understanding Data Science

Data science is the **field of study** that extracts relevant insights and develops strategy from data for business and industry.

Data science's multidisciplinary approach for analyzing **large amounts of data** blends principles and practices from **mathematics, statistics, business, machine learning, and computer engineering**.



Associated Technical Concepts

- **Statistics:** Foundation of data science, crucial for interpreting data and drawing conclusions.
- **Modeling:** Creating mathematical models to make predictions and optimize solutions.
- **Programming:** Required for implementing data science projects, with Python and R being popular choices.
- **Machine Learning:** Core to data science, enabling predictive, modeling and analysis.
- **Database:** Understanding databases to efficiently store, retrieve, and manage data.

Why Data Science is Possible in Today's World?

- *Ancient application*
  - Data science dates back to ancient world where census data, agricultural sales data, and other records to predict tax income, set prices, and anticipate flooding.
- *Why it's prominent today?*
  - Data explosion
  - Technological advancement - Computing power, data storage, advanced algorithms (AI) and softwares.

How organisations use Data Science to grow

- *Product Development*
  - Tesla uses data from its fleet of vehicles to improve the performance of its self-driving technology.
- *Customer Personalization*
  - Netflix data science to analyze viewing habits, preferences, and behaviors to recommend personalized content
- *Healthcare Research*
- By integrating genomic and environmental data, data science enhances the understanding of complex diseases like cancer, heart disease and diabetes.
- *Discussion: Share more examples of the application of data science in organisations and industries.*

Career Opportunities in Data Science

Data science is a fast-growing field that is useful different industries for variety of applications.

- **Data Scientist:** Analyzes complex data to uncover trends and insights.
- **Data Analyst:** Interprets data to support decision-making.
- **Machine Learning Engineer:** Develops algorithms and models for predictive analysis.
- **Data Engineer:** Designs and maintains data architectures and pipelines.
- **Business Intelligence Analyst:** Transforms data into actionable business insights

What a Data Scientist Does

Data Collection and Management: Data mining, web scraping, database management [Python, SQL, NoSQL].

Data Cleaning and Preprocessing: Handling missing data, data transformation, normalization [Python (Pandas, NumPy), R].

Data Analysis Exploration: Descriptive statistics, EDA, data visualization [Python (Matplotlib, Seaborn), R (ggplot2)].

Building Predictive Models: ML algorithms (regression, classification), deep learning, model evaluation [Python (Scikit-learn, TensorFlow), R, SAS].

Communicating Insights: Data visualization, storytelling with data, dashboard creation [Tableau, Power BI, Python (Plotly, Dash)].

Deploying Models: Model deployment, API integration, monitoring and maintenance.

According to Glassdoor and Forbes, demand for data scientists will increase by 28 percent by 2026, which speaks of the rising demand and need.

Glassdoor and Forbes

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THANK YOU!